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BALLARI INSTITUTE OF TECHNOLOGY & MANAGEMENT



NACC Accredited Institution*
(Recognized by Govt. of Karnataka, approved by AICTE, New Delhi & Affiliated to
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DEPARTMENT OF CSE (DATA SCIENCE)

A Mini Project Report On

“QUALITY DEFECT ANALYSIS DASHBOARD”

A report submitted in partial fulfillment of the requirements for the

DATA VISUALIZATION WITH POWER BI

Submitted By

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Visvesvaraya Technological University
Belagavi, Karnataka 2025-2026

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CERTIFICATE

This is to certify that the open-ended project report on DATA VISUALIZATION WITH POWER BI **“QUALITY DEFECT ANALYSIS DASHBOARD”** has been successfully presented by **RIDA ARSHAD** bearing USN **3BR23CD076** student of semester B.E for the partial fulfillment of the requirements for the award of **Bachelor Degree in CSE(DS)** of the BALLARI INSTITUTE OF TECHNOLOGY& MANAGEMENT, BALLARI during the academic year 2025-2026.

Signature of guide

Mr. Sundeep Sindhanoor

Signature of HOD

Dr. Aradhana D

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USN

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ABSTRACT

This project presents the development of an interactive **Quality Defect Analysis Dashboard** using Power BI to analyze manufacturing defect data. The dataset includes key variables such as product type, defect category, production units, defect weight, and production timelines. The primary objective of this project is to transform raw operational data into actionable insights that assist stakeholders in improving product quality and operational efficiency.

The dashboard leverages advanced data visualization techniques, including **Pareto analysis**, trend analysis, and categorical comparisons, to identify critical defect contributors and patterns. By highlighting the most impactful defects and their sources, the system enables decision-makers to prioritize corrective actions and optimize production processes.

Chapter 1: INTRODUCTION

1.1 Background

In modern manufacturing environments, maintaining product quality is critical for reducing costs, improving customer satisfaction, and ensuring operational efficiency. Production processes generate large volumes of defect-related data, including information about defect types, affected products, production units, and timelines.

Traditional data analysis methods are often manual, time-consuming, and lack interactivity. Business Intelligence tools like Power BI enable organizations to convert raw data into meaningful dashboards, allowing real-time monitoring and faster decision-making.

1.2 Problem Statement

Manufacturing systems often face challenges such as:

- Lack of centralized defect monitoring
- Difficulty in identifying high-impact defects
- Limited visibility into production trends
- Inefficient prioritization of corrective actions

These limitations result in delayed responses, increased production costs, and reduced product quality.

1.3 Objectives

The key objectives of this project are:

- To analyze defect distribution across products and units
 - To identify the most critical defects using Pareto analysis
 - To track defect trends over time
 - To provide a visual tool for decision-making
 - To enhance quality control through data-driven insights
-

1.4 Scope

The dashboard is designed for:

- Quality control engineers
- Production managers
- Operations teams
- Business analysts

It can be extended to various manufacturing industries where defect monitoring is essential.

Chapter 2: DATASET DESCRIPTION

The dataset used in this project is a structured manufacturing dataset containing multiple attributes related to production and defects.

2.1 Key Fields

- **Product:** Represents different product categories (e.g., GI, PPGL, GL)
- **Defect:** Type of defect observed during production
- **FinalWt:** Weight associated with the defect (impact measure)
- **Unit:** Production unit or machine identifier
- **ProductionEndDate:** Date of production completion
- **Additional Fields:**
 - MachineID
 - OperatorID
 - ProcessStage
 - NCOFlag (indicates non-conformance)

2.2 Data Characteristics

- Contains thousands of records representing production output

- Includes both categorical and numerical variables
- Time-based data enables trend analysis
- Multiple defect types (high dimensionality)

Chapter 3: TOOLS & TECHNOLOGIES USED

- **Power BI Desktop:** Used for data visualization and dashboard creation
- **Microsoft Excel / CSV:** Used as the primary data source
- **DAX (Data Analysis Expressions):** Used to create calculated measures and logic
- **Data Modeling Techniques:** Used to structure data for efficient analysis

Chapter 4: DATA MODELLING

The dataset follows a **single-table model**, allowing direct aggregation and filtering.

Analytical Relationships:

- Product → Defect Contribution
- Unit → Defect Distribution
- Date → Trend Analysis
- Defect → Pareto Ranking

This structure enables flexible slicing and aggregation without requiring complex joins.

Chapter 5: DATA PREPROCESSING

Several preprocessing steps were performed to ensure data quality:

- Removal of null and missing values
- Replacement of “unknown” values with nulls
- Correction of data types (dates, numeric fields)
- Standardization of defect naming conventions
- Filtering irrelevant time periods

- Validation of numerical consistency

These steps ensured accurate and reliable analysis.

Chapter 6: DAX MEASURE CREATED

The following measures were created using DAX:

Basic KPIs

- **Total Weight**

Total Weight = SUM(data[FinalWt])

- **Total Records**

Total Records = COUNTROWS(data)

- **Unique Defects**

Unique Defects = DISTINCTCOUNT(data[Defect])

Advanced Measures

- **Defect Rank (for Pareto)**
- **Cumulative Weight**
- **Pareto %**

These measures enable identification of top defect contributors and cumulative impact.

Chapter 7: DASHBOARD VISUALIZATIONS

The dashboard consists of multiple interactive visuals:

1. KPI Cards

Provide a quick summary of key metrics:

- Total Defect Weight (81.09K)

- Total Records (8K)
 - Unique Defects (96)
-

2. Pie Chart — Defect Contribution by Product

Shows how different products contribute to total defects.

3. Unit-wise Defect Distribution

Displays defect contribution across production units.

4. Bar Chart — Top Defects by Impact

Ranks defects based on their contribution to total defect weight.

5. Line Chart — Defect Trend Over Time

Shows how defects vary across production dates.

6. Pareto Chart — Defect Analysis

Combines bar and line chart to highlight critical defects contributing to majority of issues (80/20 rule).

Chapter 8: DASHBOARD INSIGHTS

The dashboard provides several key insights:

- A small number of defects contribute to the majority of total defect weight
 - GI product has the highest defect contribution
 - Certain units (e.g., CRM1CGL3) show higher defect concentration
 - Defect trends indicate a sharp increase in later production stages
 - Top defects such as scratches and coating issues dominate overall impact
-

Chapter 9: BENEFITS OF THE SYSTEM

- Enables faster identification of critical defects

- Improves decision-making through visual insights
 - Reduces manual data analysis effort
 - Enhances production quality monitoring
 - Supports continuous improvement initiatives
-

Chapter 10: LIMITATIONS

- Dataset is static (no real-time integration)
 - No predictive analytics implemented
 - Limited root cause analysis
 - No automated alerts or notifications
-

Chapter 11: RESULTS

The dashboard successfully transforms raw defect data into a centralized analytical platform.

Key Results

- Total defect weight: **81.09K**
- Total records analyzed: **8,000+**
- Unique defect types: **96**
- Top 20% defects contribute to majority of total defect weight
- Significant increase in defect occurrence observed in later production timeline

81.09K

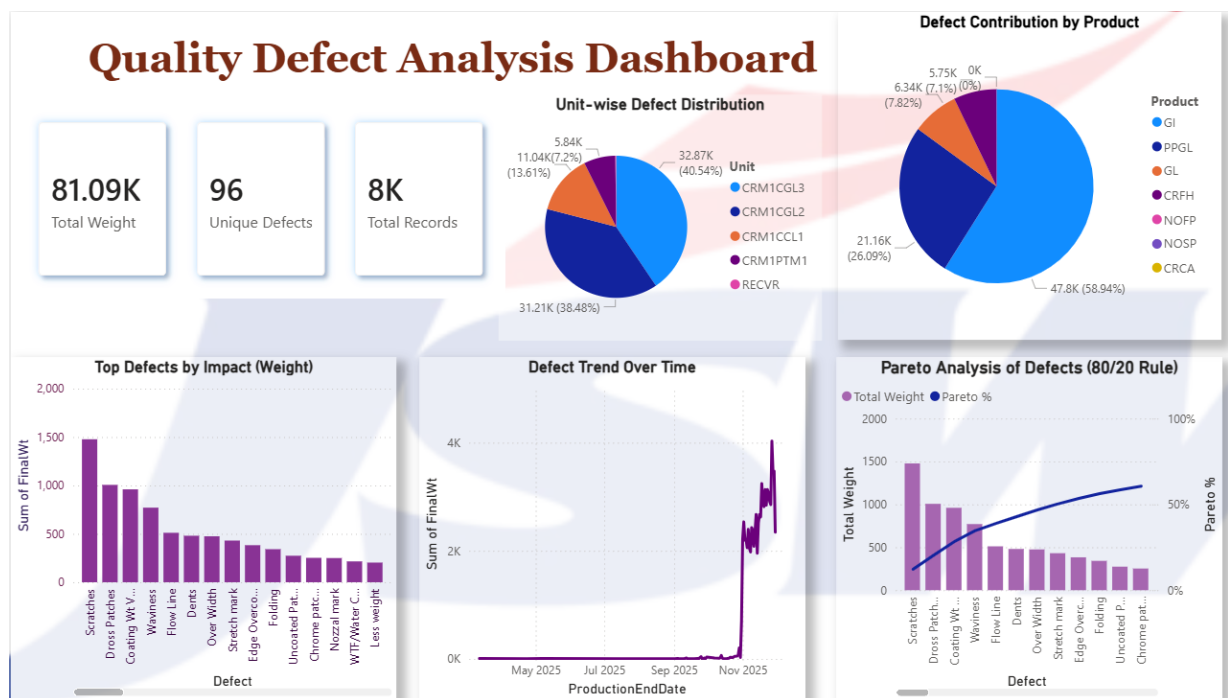
Total Weight

96

Unique Defects

8K

Total Records



Chapter 12: CONCLUSION

This project demonstrates the effectiveness of Power BI in analyzing manufacturing defect data. By integrating data visualization with analytical techniques such as Pareto analysis, the dashboard provides valuable insights into defect patterns and operational performance.

The system helps organizations focus on high-impact issues, improve quality control, and make informed decisions.